Precept 1: Warm Up Exercise

September 9, 2026

Problem

Earlier in class we defined the projection of a vector $\vec{v}$ on to $\vec{u}$ as

$$\text{proj}_{\vec{u}}(\vec{v}) = \frac{\vec{v}\cdot\vec{u}}{\vec{u}\cdot\vec{u}}\vec{u} \tag{1}$$

But really, we don't need to define it, we can derive it!

Derive the formula for $\text{proj}_{\vec{u}}(\vec{v})$ using vector addition, scalar multiplication, the dot product, orthogonality.

Hint: The following expressions will be helpful:

$$(\vec{x}+\vec{y})\cdot\vec{z} = \vec{x}\cdot\vec{z} + \vec{y}\cdot\vec{z} \qquad \text{and} \qquad (c\vec{x})\cdot\vec{y} = c(\vec{x}\cdot\vec{y})$$

where c is a scalar.

Solution

We want to decompose $\vec{v}$ into

$$\vec{v} = c\vec{u} + \vec{w} \tag{2}$$

where $\vec{u}$ and $\vec{w}$ are orthogonal

$$\vec{u}\cdot\vec{w} = 0$$

Our task is to find the coefficient c. By taking the dot product with $\vec{u}$ on either side of (2), we can write

$$\begin{aligned}\vec{v}\cdot\vec{u} &= (c\vec{u}+\vec{w})\cdot\vec{u}\\ &= (c\vec{u})\cdot\vec{u} + \vec{w}\cdot\vec{u}\\ &= c(\vec{u}\cdot\vec{u}) + \vec{w}\cdot\vec{u}\end{aligned}$$

Notice that $\vec{w}\cdot\vec{u} = 0$ by construction and we can rearrange things to write

$$c = \frac{\vec{v}\cdot\vec{u}}{\vec{u}\cdot\vec{u}}$$

where our projection vector is now

$$c\vec{u} = \frac{\vec{v}\cdot\vec{u}}{\vec{u}\cdot\vec{u}}\vec{u}$$

which was to be shown.